

**Ninth Semester Project
Report**

Modeling Clouds in Planetary Atmospheres



Submitted by

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Declaration

I do hereby declare that the project report entitled "Modelling Clouds in Planetary Atmosphere" was carried out by me under the guidance of my supervisors Dr. Luke Chamandy, SPS NISER, and Dr. Jayesh M Goyal, SEPS, NISER.

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Abstract

The clouds on Jupiter are predominantly composed of ammonia (NH_3), as the planet's atmospheric temperature and pressure conditions favor its condensation.

In this work, I have implemented a numerical model to calculate the vertical profiles of particle size distributions for condensate species in Jovian atmospheres. The methodology closely follows the framework of Ackerman and Marley (2001)[1], employing a constant sedimentation efficiency parameter (f_{sed}) to represent the balance between upward turbulent mixing and downward sedimentation. This approach allows for meaningful comparison with the earlier models of Lewis, Marley et al., and Lunine et al. By computing ammonia cloud profiles for varying sedimentation efficiencies, I investigate how cloud particle properties and vertical structures evolve across different atmospheric pressure levels. The resulting model serves as a foundation for extending cloud formation simulations to exoplanetary atmospheres using our group's in-house adaptive planetary atmospheric model, **SANSAR**.

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Introduction

Clouds play a fundamental role in regulating the radiative and dynamical structure of planetary atmospheres. They shape both reflected and emitted spectra by absorbing and scattering radiation, thereby influencing a planet's albedo and thermal balance. A persistent challenge in atmospheric retrievals is the uncertainty in the vertical location and structure of clouds on distant planets, which complicates the interpretation of observed spectra.

On Jupiter, the uppermost cloud deck forms when ammonia (NH_3) reaches saturation and condenses at pressures of roughly 0.1 - 0.42 bar. The resulting cloud structure depends on the balance between condensation, vertical mixing, and particle sedimentation. The widely used[1] framework provides a one-dimensional description of this equilibrium by introducing the sedimentation efficiency parameter f_{rain} which measures the ratio of downward settling velocity to upward turbulent mixing. Varying f_{rain} allows the model to represent a spectrum of cloud morphologies—from vertically extended decks with small particles to thin, compact clouds with larger, fast-settling particles.

In this work, I implement the[1] approach to compute vertical profiles of condensate mixing ratios and particle sizes for ammonia clouds in Jupiter's atmosphere. By exploring different f_{rain} values, I analyze how sedimentation efficiency controls cloud thickness and particle growth. These results provide a foundation for incorporating cloud microphysics into exoplanet atmosphere modeling within our group's **SANSAR** framework.

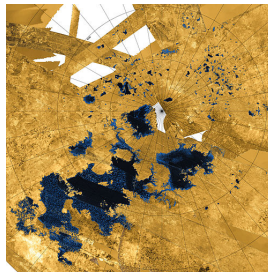


Figure 1: Picture of methane lakes on the surface of Titan (Saturn's Moon) taken by the Cassini Spacecraft indicating that the atmosphere consists largely of methane clouds

Chapter 1

Clouds

1.1 The fundamentals of cloud formation

In meteorology, clouds are suspensions of liquid droplets, ice particles, condensate aerosols, or other particles suspended in the atmosphere of a planetary body or similar space. Water or various other chemicals may be comprising as crystals and droplets, e.g.- ammonia clouds on Jupiter and water vapor clouds on Earth.

In planetary atmospheres—especially giant planets such as Jupiter—clouds exert a dominant control on radiative transfer, atmospheric circulation, thermal structure, and observable spectra. Their formation and evolution arise from a balance between thermodynamics, micro-physical growth processes, and turbulent transport.

Cloud microphysics deals with the small-scale physical processes that govern the formation, growth, and movement of cloud particles. These include condensation, nucleation, evaporation, sedimentation, and turbulent mixing. Although these processes occur on microscopic scales, their cumulative effects determine large-scale cloud structure and optical properties.

1.1.1 Cloud Formation

Air Parcel and Turbulence

In any planetary atmosphere, molecular mixing of gases and substances takes place only up to a few centimeters in the atmosphere. To talk about mixing of substances across the vertical coordinate, in macroscale, the concept of air parcels is important. Each air parcel is assumed to be :

- i) thermodynamically insulated from its environment, and its temperature changes adiabatically as it rises or sinks and always remains at the same pressure level as the environmental air in the surroundings, which is assumed to be in hydrostatic equilibrium.
- ii) moving slowly enough that the kinetic energy of the air parcel is at a negligible scale of the total energy.

[2] When this warm air parcel laden with the gaseous form of elements and chemical species rises, it expands and cools. If the chemical species cools to a temperature where it can condense, a condensation cloud will form. The moment the temperature of the air parcel is equal to the **dew point temperature** (where relative humidity is 100 %) of the species, the cloud base is formed. Now the parcel of air will continue rising up to high altitude levels until the surrounding atmospheric air becomes cooler than the air parcel, and that is where the cloud formation stops. This vertical length till which the cloud is formed is called the height of the cloud. A species can only start to condense when the partial pressure of the species vapor exceeds the partial pressure of the gaseous species when air is fully saturated. For any gaseous species that might condensate,[2]

$$P_g(T) = X_g P(T) \geq P_{Vg}(T)$$

where $P(T)$ is the total gas pressure, P_g is the gas partial pressure, X_g is the gas mixing ratio and P_{Vg} is the saturation vapor pressure.

The cloud base is at the intersection of the planet's temperature-pressure profile with the saturation vapor pressure curve. This curve is the equilibrium curve between the gas and liquid or solid state of the species.

The saturation vapor pressure as a function of temperature is known as the Clausius Clapeyron Curve - [2]

$$\frac{dP_v}{dT} = \frac{L}{T(V_2 - V_1)}$$

Where $V = 1/\rho$ and is known as the specific volume and 1,2 subscript refers to the transition of state of species from liquid or solid to gaseous phase. If we consider the ideal gas then the specific volume of solid or liquid is small compared to the vapor phase, i.e $(V_2 - V_1) \sim V_1 = R_v T/P$ where R_v is the specific gas constant for the vapor in the planet. Then the above equation becomes :

$$\frac{dP_v}{dT} = \frac{LP_v}{R_v T^2}$$

here L is latent heat of vaporisation. As the air parcel rises it expands and cools adiabatically until $P_v = P_{sat}$. Beyond this point the air becomes supersaturated, and condensation begins. In many cloud models (here, [1] the assumption is made that supersaturation is negligible due to abundance of condensation nuclei. Thus,

$$q_s = \frac{R_d}{R_v} \frac{P_{sat}}{P - P_{sat}}$$

where P is the total pressure and R_d is mass of dry air in the planet. Now, for clouds to form in a large scale height and be broad, only molecular diffusion is not enough. We need the intervention of turbulence. The large scale mixing is mostly facilitated by the set up of convective cycles or convective current in which the hot air parcel

risers passing the dew point temperature and beginning the formation of clouds the hot air parcel rises and cools down and while the heavier, colder condensed species is coming down it is again getting re-evaporated due to the surrounding atmospheric temperature being warmer than it and thus a cloud is sustained. To sustain a cloud, molecular mixing, turbulence mixing and convection current must always be an active process . Only when the condensed species is way cooler than the surrounding atmospheric layer do we see precipitation in form of rain, snow or virga. Turbulence continuously mixes vapor and condensate vertically. Vertical mixing is often parameterized as an eddy diffusion process:

$$F_{mix} = -K_{zz} \frac{dq_t}{dz}$$

here, K_{zz} is the 1-D eddy diffusion coefficient along the vertical coordinate q_t is the total mass mixing ratio (vapor (q_v) + condensate (q_c)). This term behaves familiar to **Fick's first law of diffusion** transporting vapor upward from deep levels and lifting condensate upward against gravity.

In the convective regions, K_{zz} is derived from the mixing length theory. A

Sedimentation (Rainout)

This is another factor which comes into play when formation of clouds. The heavier condensate particles sink to the bottom layers of the cloud due to action of gravity and thus resulting in rainout or precipitation. For turbulent flow and laminar flow of the droplets we use the stokes flow in low Reynolds number regime and other methods (emperical drag law) to fit the droplet size of the particles in their fall radius categories. To give a general idea, the fall radius of a particle is associated with the fall velocity with :

$$v_{fall} = \frac{2}{9} \frac{\beta(\rho_p - \rho_a)gr^2}{\eta}$$

η is the dynamic viscosity, g is gravitational acceleration, ρ_p is density of condensate , ρ_a is density of atmosphere , r is the droplet radius and β is the Cunningham slip factor . B[1]

1.1.2 Ackerman Marley Cloud Model

[1] provide a 1-D steady-state cloud model where sedimentation and mixing are in balance. This is a very simple yet effective model to describe the dynamics of cloud formation effectively.

The equation :

$$-K_{zz} \frac{\delta q_t}{\delta t} - f_{rain} w_* q_c = 0 \quad (1.1)$$

here,

- K_{zz} is eddy diffusion coefficient
- $\frac{\delta q_t}{\delta t}$ is the variation of total ($q_c + q_v$) mass mixing ratio
- f_{rain} is the rainout parameter (ratio of mass weighted droplet sedimentation velocity (v_{fall}) to convective velocity (w_*))
- w_* is the convective velocity scale
- q_c condensate mass mixing ratio

For analytical derivations refer to **B**

1.2 Method

Approach to the Modeling

We are using python interface to model this problem. We first created the pressure temperature layers by making a grid and named it function . Our data was taken from the data capture by the spacecraft **Voyager** and hence it is a realistic data. The function eddysed in the code loops through the gas condensate ammonia (we will options to add more gases to the analysis later) while calling the function of layer. Then it calls the function $calc_{qc}$ to each of the layers of (P,T)of the data from voyager and calculates some basic quantities like mean free path, number density , atmospheric viscosity which are required for the calculations of the droplet size radius of the particles of ammonia. In the function $calc_{qc}$ we start by computing the saturation vapor pressure P_{sat} at a given layer i that has its own corresponding pressure and temperature (T_i and P_i) . It is computed through the formula given in **A**. Then it is used to calculate the saturation vapor mass mixing ratio through the formula

$$q_s = \frac{f_s P_{sat}}{R_c T_i \rho_a}$$

where $f_s = S+1$ is the super saturation which is assumed to be 1 in this model (i.e., S is user defined as 0 it corresponds to potential supersaturation prior to condensation) and R_c is the universal gas constant. q_v (vapor mass mixing ratio) is then compared to q_t in the immediately preceding layer q_{below} . If $q_{below} > q_v$ then we proceed to solve 1.1 numerically through the methods. However to analytically solve the equation 1.1 is the form of :Refer to **B**

$$q_{t,i} = q_{c,i} + (q_{below} - q_{v,i}) \exp(-f_{sed} * \Delta z / L) \quad (1.2)$$

Here, L is the mixing length whose calculation is described in the **A**. This method is the numerical method and here q_v is assumed to be constant throughout the atmospheric layers. In reality it is not the case and we will be taking q_v from equilibrium chemistry in the later stages of the development of the model. We then proceed to calculate the particle fall radius v_{fall} from $calc_{qc}$ and we tried to use

the **root finding algorithm - secant method** and the **brentq** Method (see **A** for definitions) to find the values for the fall radius (r_w) of the particle. However we are still facing some convergence issues and the problem is still in debugging stage. f_{rain} is a user defined parameter and we have used it to benchmark our model the figure 1. of [1]). We stress on the fact that $w_* = K_{zz}/L$ and the method we used to calculate K_{zz} is detailed in the paper of [3]. More details in the **(A)**.

From the particle fall radius r_w we want to obtain the geometric mean radius r_g via the following equation :

$$r_{g,i} = r_{w,i} f_{sed}^{1/\alpha_1} \exp\left(-\frac{\alpha_i + 6}{2} \ln^2 \sigma_g\right) \quad (1.3)$$

with the associated column droplet number concentration (cm^{-2}) :

$$N_{dz_i} = \frac{3\rho_{a,i} q_{c,i}}{4\pi\rho_p r_{g,i}^3} \exp\left(-\frac{9}{2} \ln^2 \sigma_g\right) dz_i \quad (1.4)$$

where α is the power-law dependence for the particle fall-speed about the solution to the following equation :

$$v_f(r_w) - w_* = 0 \quad (1.5)$$

The fall particle radius is that radius which is computed by finding the radius which balances of the fall speed of the particle (assumed to be spherical here) and the convective velocity scale.

Calculation of the mass mixing ratios

Here q_c , the condensate mixing ratio is calculated using the formula $q_c = q_t - q_s$ and not $q_c = q_t - q_v$ because here, when the cloud starts to form the value of the q_v is capped to q_s as q_s is the maximum vapor mixing ratio that can be held in equilibrium with the condensate at local T,P. To make it more simple , here is how q_t, q_v, q_s, q_c play their roles throughout the atmosphere

1. **Below base of the cloud** $q_v = q_t, q_c = 0, q_v < q_s$
2. **At cloud base** (condensation starts) $q_v = q_s, q_t = q_s, q_c = 0$
3. **Above cloud base** $q_v \rightarrow q_s, q_c = q_t - q_s$

Here the *Lewis cloud model* (1969)[4] assumption is made and we are starting below the cloud base where $q_c = 0$ and $q_v = q_{below}$ and condensation of all vapor in excess of saturation is happening at each consecutive layer upward.[1] Thus,

$$q_c(z) = \max[0, q_v(z - \Delta z) - q_s(z)] \quad (1.6)$$

$$q_v(z) = \min[q_v(z - \Delta z), q_s(z)] \quad (1.7)$$

1.3 Physical Intuition behind the equations

This one- dimensional Eulerian framework of [1] talks about the balance of the two opposite forces - the convective turbulent mixing force acting upwards while maintaining a constant vapor mixing ratio below the cloud and the sedimentation force bringing the condensate particle downward due to action of gravity. Below the cloud base air is too warm and thus saturation vapor pressure is large and so vapor cannot reach saturation hence we have no condensation. At the cloud base. Vapor mixing ratio equals saturation and condensation begins. Above the cloud base The air becomes supersaturated unless condensation removes vapor and because condensation happens fast relative to atmospheric mixing thus $q_v \rightarrow q_s$ (quasi saturated) and excess vapor condenses, q_c increases with increase in altitude. It justifies one of the assumptions that supersaturation is 0 in this model and hence $q_v = q_s$ as the atmosphere is assumed to be nearly saturated . Allowing the supersaturation that persists even after condensation is accounted for represents conditions in which condensation is too slow or there are significant barriers to droplet formation.

Effect of supersaturation :

1. Below cloud base : Pressure and Temperature is high and thus saturation vapor mixing ratio is large if the supersaturation is increased then the threshold rises and the atmosphere can remain unsaturated for longer period of time, thereby pushing the cloud base upward.
2. At cloud base : when the total mixing ratio is equal to the saturation mixing ratio so when the supersaturation is increased then cloud starts forming at higher altitudes and cooler temperatures.
3. Above cloud base : the total mass mixing ratio exceeds the saturation mixing ratio due to condensation and if supersaturation is increased then q_c becomes smaller, clouds become less denser and droplet sizes become smaller and the sedimentation slows down due smaller droplet size, thereby giving a more extended yet less opaque cloud.

Effect of w_* :

1. Larger value of w_* leads to faster mixing, leading turbulent mixing to dominate
2. Smaller value of w_* leads to slower mixing, leading sedimentation to dominate

Effect of f_{rain} :

1. smaller value of f_{rain} = vertically thick clouds, extending through atmosphere leading upward transport dominate
2. Larger value of f_{rain} = thinner depleting clouds leading sedimentation dominate

1.4 Result and Discussion

Upon solving the equation and comparing our graph with the Figure of [1], for benchmarking of my model, we found that our cloud base is as 0.379 bar while there is at 0.42 bar for the graphs plotted for various values of $f_{rain} = (0.5, 1, 3, 10)$. On the X- axis lower axis, we have $\log_{10} q_c$, and on the x upper axis we have temperature in Kelvins and on the Y axis we have pressure(bar) in the log scale ranging from 1 bar to 0.1 bar.

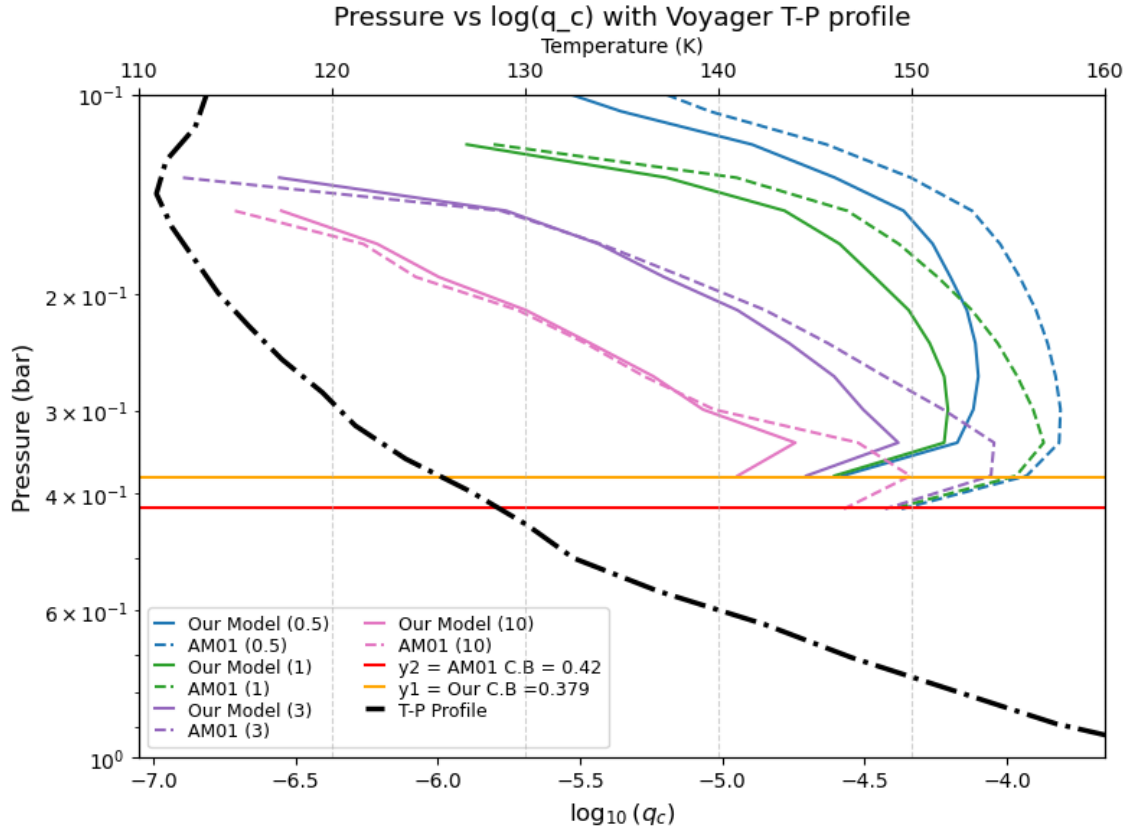


Figure 1.1: Pressure vs $\log_{10} q_c$ for various f_{rain} values and compared with Figure : 1 of [1]

As we can see that for lower value of f_{rain} we have convergence issue and the cloud base is uniformly shifted for all the calculations of f_{rain} to 0.379 while the model simulation of [1] is having the cloud base at 0.42 bar. The reason for it could be precision issue and the way the indexing is happening in the Fortran code (original version of their code is in Fortran) is different from the way it is encountered in the Python language. However, I judged that the issue is with the *root finding algorithm* being dealt with in python and switched to *Brent* method however the result remained unchanged and we got same values for r_w and r_g

which is not physically and mathematically possible (that would imply that all the droplets are of the same size).

I am currently unable to figure out the exact problem with the implementation of the code though the search and debugging is still on.

But as one can see, at higher values of f_{rain} (3,10) the clouds are almost matching the trend in the upper layers of the atmosphere though the shift of the cloud base cannot be neglected. For lower values of f_{rain} (0.5,1) the graphs are differing a lot from the actual model though the trends and curvature look almost similar to the ones of the model. I have also carefully checked the unit conversions and found no problem there as well. At the moment I can outline the reason for this anomaly to precision issue and convergence issue.

However from the analysis of the graph we can see that for larger values of f_{rain} the cloud we are getting thinner clouds due to low value of q_c and for lower values of f_{rain} we are getting thicker and dense clouds because of higher values of q_c which condensate at relatively higher levels of the atmosphere than the ones with higher f_{rain} value. We also tried the analytical method to solve the problem but we were still facing issues of convergence in the higher f_{rain} values. The graphs of comparison with [5] are in B.

Conclusion

In this semester we tried to recreate the *Ackerman and Marley, 2001* cloud model for the ammonia clouds of Jupiter. While our attempt to write the code for this formulation in python language is not entirely complete but it still has some progress in it. While we are differing at the cloud base and the solution of the differential equation mostly at the lower values of f_{rain} , we are still working to find the plausible causes for this issue.

The first term in the differential equation corresponds to the diffusion gradient term talking about the turbulent mixing of the convective current upwards and the second term talks about the sedimentation term which brings down the condensate due to action of gravity which together with the fall radius of the particles r_w puts a constrain on q_c .

The analysis of the results from our graph (even though shifted is still consistent with the theory) load bases forming near the temperature–pressure level where $q_v = q_s$, condensate mass increasing above this point, and particle sizes responding sensitively to f_{rain} . Higher sedimentation efficiency produces larger particles, thinner clouds, and reduced condensate mass, while low f_{rain} yields small particles and extended cloud decks.

So far our qualitative analysis is correct with what was predicted in the model but the quantitative constraints are yet to be improved and determined well as we observe a converge issue for the lower f_{rain} values but a similarity in the trend and shape of the graph. While both *root finding algorithm* and *Brent* method is giving the similar trend of the graph in our case. The convergence of the solution for the Brent method is better than that of the root finding algorithm.

The reconstruction of this model not only improves our understanding of Jovian cloud microphysics, but also provides a robust foundation for extending cloud formation calculations to exo-planetary atmospheres. The methods developed here—root-finding, supersaturation control, fall-speed calculations, and vertical transport—will directly feed into our group’s adaptive atmospheric model **SANSAR**, enabling consistent cloud predictions for diverse planetary environments.

Appendix A

Some basic definitions and formulas

- **Hydrostatic equilibrium**- $p(z) = \int_z^\infty g\rho dz$, where height is z , ρ is the density of the air, g is the gravitational acceleration of the planet.
- **Dew point temperature** - the temperature to which surrounding atmosphere must be cooled, at constant pressure and moisture content, in order for it to become fully saturated with the condensate species.
- **Condensate** - a liquid that results from the cooling and condensation of a gaseous species
- q_c - moles of condensate per moles of atmosphere
- q_v - moles of vapor per moles of atmosphere
- q_t - sum of moles of (condensate+vapor) per moles of atmosphere
- q_s - moles of vapor per moles of atmosphere when atmosphere is saturated
- $R_d = (1000 \times \frac{8.3145}{2.2} JK^{-1}kg^{-1})$, where 8.3145 is the universal gas constant and 2.2 is the molecular weight of dry air in **Hydrogen-Helium** dominated atmosphere of Jupiter
- $R_v = (1000 \times \frac{8.3145}{18.016} JK^{-1}kg^{-1})$, where 8.3145 is the universal gas constant and 18.016 is the molecular weight of water
- **Convective Current** - a circulating flow within a fluid (liquid or gas) that occurs when differences in temperature cause corresponding differences in density of the particles.
- **Lapse Rate** (Γ_i) - the rate at which temperature changes with an increase in altitude.

- **Dry Adiabatic Lapse Rate** ($\Gamma_{adiabatic}$) - $\frac{g}{c_p}$, g is gravitational acceleration of the planet and c_p is the specific heat constant at constant pressure
- **Mixing length theory**- A common local first order closure of the turbulence closure problem. It is analogous to molecular diffusion, it assumes that flux is linearly proportional to and directed down the local gradient. $F_H = -K \frac{\delta \theta}{dz}$
- **Eddy diffusion coefficient**- a parameter used instead of molecular diffusivity and is a constant of proportionality which increases with the intensity of the turbulence, it varies by height above the ground, mean wind shear and surface heating by the sun through radiation. Formula : [3]

$$K_{zz} = \frac{H}{3} \left(\frac{L}{H} \right)^{\frac{4}{3}} \left(\frac{R F_{conv}}{\mu \rho_a c_p} \right)^{\frac{1}{3}}$$

where $H = RT/\mu g$, μ is atmospheric molecular weight, T is temperature, R is universal gas constant, L is turbulent mixing length, $F_{conv} = \sigma T_{eff}^4$ where σ is Stefan - Boltzmann Constant, T_{eff}^4 is the temperature at the surface of the planet, ρ_a is atmospheric density, c_p specific heat of the atmosphere.

- **Mixing Length** - The average size of the length of eddies formed during turbulent mixing length. Formula:

$$L = H \times \frac{\Gamma_i}{\Gamma_{adiabatic}}$$

where H is the scale height

- **Convective Velocity scale** (w_*) - a velocity scale which is useful to characterize the turbulent mixing due to free convection in an unstable stratified atmosphere.
- **Root Finding Algorithm** - Iterative method for finding the roots in a continuous equation.
- **Secant Method** - Replacing the method of Newton to find a continuous derivative of the function with a finite difference gives rise to secant method.
- **Brent's Method** - A compilation of three techniques to achieve the root of a function - secant method, bisection and inverse quadratic interpolation
- **Formula to calculate Saturation Vapor pressure (P_{sat}) for Ammonia :**
 $e_s = \exp(10.53 - \frac{2161}{T} - \frac{86596}{T^2})$ where e_s is in dynes cm^{-2}

Appendix B

Some Derivations

B.1 Analytical derivation of the cloud formation equation

here $K_{zz} = K$

$$-K \frac{dq_t}{dz} - f_{rain} w_* q_c = 0 \quad (B.1)$$

$$-K \frac{dq_t}{dz} = f_{rain} w_* q_c \quad (B.2)$$

$$\frac{-K \frac{dq_t}{dz}}{q_c} = f_{rain} w_* \quad (B.3)$$

$$-K \int \frac{1}{q_c} \frac{dq_t}{dz} dz = z f_{rain} w_* \quad (B.4)$$

integrating it over altitude

$$-K \int_{L_1}^{L_2} \frac{dq_t}{q_t - q_v} = z f_{rain} w_* \quad (B.5)$$

$$-K \log(q_t - q_v)|_{L_1}^{L_2} = z|_{L_1}^{L_2} f_{rain} w_* \quad (B.6)$$

now we know : $w_* = K_{zz}/L$ putting it in the equation and changing the indexing of q_t & q_c for L_1 & L_2 :

$$-k[\log(q_{t,i} - q_{v,i}) - \log(q_{below} - q_{v,i})] = \Delta z f_{rain} \frac{K}{L} \quad (B.7)$$

$$\log \frac{(q_{t,i} - q_{v,i})}{(q_{below} - q_{v,i})} = -f_{rain} \frac{\Delta z}{L} \quad (B.8)$$

This is the analytical solution of the equation.

B.2 Similarity with other physical equations:

1st Term Fick's 1st law of Diffusion is analogous to the first term of the equation of cloud model:

Fick's first law of diffusion:

$$J = -D \frac{dC}{dx} \quad (B.9)$$

J = diffusive flux (amount of substance crossing unit area per unit time)

D = diffusion coefficient

C = concentration of the substance

x = spatial coordinate

$-$ = negative sign shows flux goes down the concentration gradient

It is analogous to the first term of the equation $-K \frac{dq_t}{dz} - f_{rain} w_* q_c = 0$ in:

$$T_1 = -K \frac{dq_t}{dz} \quad (B.10)$$

where , K is analogous to D - it is the eddy diffusion coefficient

q_t is analogous to C - it is total mass mixing ratio

here $x = z$, which is the vertical coordinate

2nd Term

The second term is analogous to the sedimentation flux. Sedimentation flux is defined as : the rate at which sediment is transported across a specific area, such as an atmosphere. It is calculated by measuring the concentration of suspended sediment and the velocity of the fluid carrying it.

Therefore sedimentation flux will be:

$$F_{sed} = f_{rain} v_{fall} q_c \quad (B.11)$$

where :

q_c = condensate mass mixing ratio (g condensate / g air)

v_{fall} = particle terminal velocity (cm/s)

Now, in our equation $f_{rain} = \frac{v_{fall}}{w_*}$

Thus, $v_{fall} = f_{rain} \times w_*$ So, comparing to our second term of the main equation

$$f_{rain} w_* q_c = f_{rain} v_{fall} \quad (B.12)$$

Similar to real life physical equation

our equation

$$-K \frac{dq_t}{dz} - f_{rain} w_* q_c = 0$$

is similar to charged particle transport.

$$-D \frac{dn}{dx} + \mu n E$$

where D is charged diffusivity, μ mobility of ions, n is the concentration, E is the electric field, x is the spatial coordinate.

B.3 Reynolds number and Droplet size distribution

In the atmosphere apart from turbulent flow there is also laminar flow which depends on the temperature, density of atmospheric (which is assumed to be uniform in this case) mass and diameter of the particle in consideration along with velocity of the particle falling. Reynolds number comes here in play as it tells us whether the particle is going through laminar flow or dynamic flow.

$$R_e = \frac{\rho v L}{\eta} \quad (B.13)$$

where, ρ - density, v - speed of the particle, L - characteristic length scale, η - dynamic viscosity.

$$\eta = \frac{5}{16} \frac{\sqrt{\pi m k_B T} (k_B T / \epsilon)^{0.16}}{\pi d^2 \times 1.22} \quad (B.14)$$

here, d is molecular diameter, k_B is Boltzmann constant, m is the mass of the particles, ϵ is the depth of the Leonard-Jones potential well for the atmosphere [1]

In our model we have defined the Reynolds number as :

$$2r \frac{\rho_a v_{fall}}{\eta} \quad (B.15)$$

here, r is the radius of the droplet, v_{fall} is calculated according to the distinction of the Reynolds number'

For Reynolds number less than 1

Over here we have laminar flow;

$$v_{fall} = \frac{2\beta g r^2 \Delta \rho}{9\eta} \quad (B.16)$$

here, $\beta = (1 + 1.26 K_N)$ accounts for gas kinetic effects is the Cunningham correction factor for non-continuum effects, in which Knudsen Number N_{KN} is the ratio of the molecular free path to the droplet radius $N_{KN} = \frac{\lambda}{r}$ and $\Delta \rho$ is difference of densities of the condensate and the atmosphere.

For Reynolds number between 1 and 1000

For Reynolds number in this category we have intermediate turbulent flow;

$$C_d R_e^2 = \frac{32r^3 \rho_a (\Delta \rho)}{3\eta^2} \quad (\text{B.17})$$

here C_d is the drag coefficient, which is used in drag force

$$F_\eta = \frac{C_d}{2} \pi \rho_a r^2 v_f^2$$

[5] Now, $C_d R_e^2$ is independent of the fall velocity and hence we fit $y = \log(R_e)$ as a function of $x = \log(C_d R_e^2)$. The fit goes as $y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + b_4 x^4 + b_5 x^5 + b_6 x^6$ where $b_0, b_1, b_2, b_3, b_4, b_5, b_6$ are coefficients of the fit.

For Reynolds number greater than 1000

Here the flow is completely turbulent. So, we assume the drag coefficient to have a fixed asymptotic value $C_d = 0.45$ [1] which gives us

$$v_{fall} = \beta \sqrt{\frac{8gr\Delta\rho}{3C_d\rho_a}} \quad (\text{B.18})$$

B.3.1 Calculation of Effective droplet radius

The geometric mean radius r_g via the equation 1.3:

with the associated column droplet number concentration (cm^{-2}) via the equation 1.4 :

Then , the effective droplet radius is calculated as :

$$r_{eff,i} = r_{g,i} \exp\left(\frac{5}{2} \ln^2 \sigma_g\right) \quad (\text{B.19})$$

here $\sigma_g = 2$, is the geometric standard deviation, it can lie between 1.1 - 2.5

B.4 Comparison with Virga

We have compared our model and [1] figure 1 with Virga [5] and found the following graphs:

The comparison show that Virga is using a different solving method and precision factor to solve for q_c values for various f_{rain} of the clouds . In its comparison my model and [1] looks a lot similar. These graphs were added for a qualitative understanding with various models.

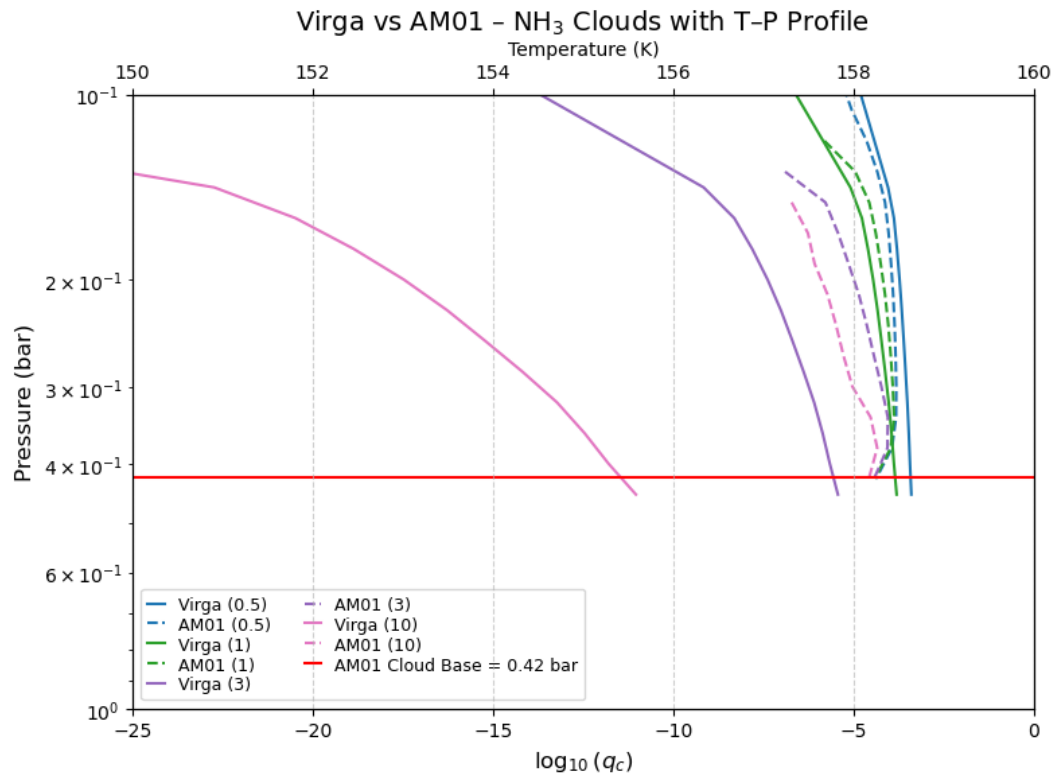


Figure B.1: Pressure vs $\log_{10}q_c$ for various f_{rain} values and compared with Figure : 1 of [1] and [5]

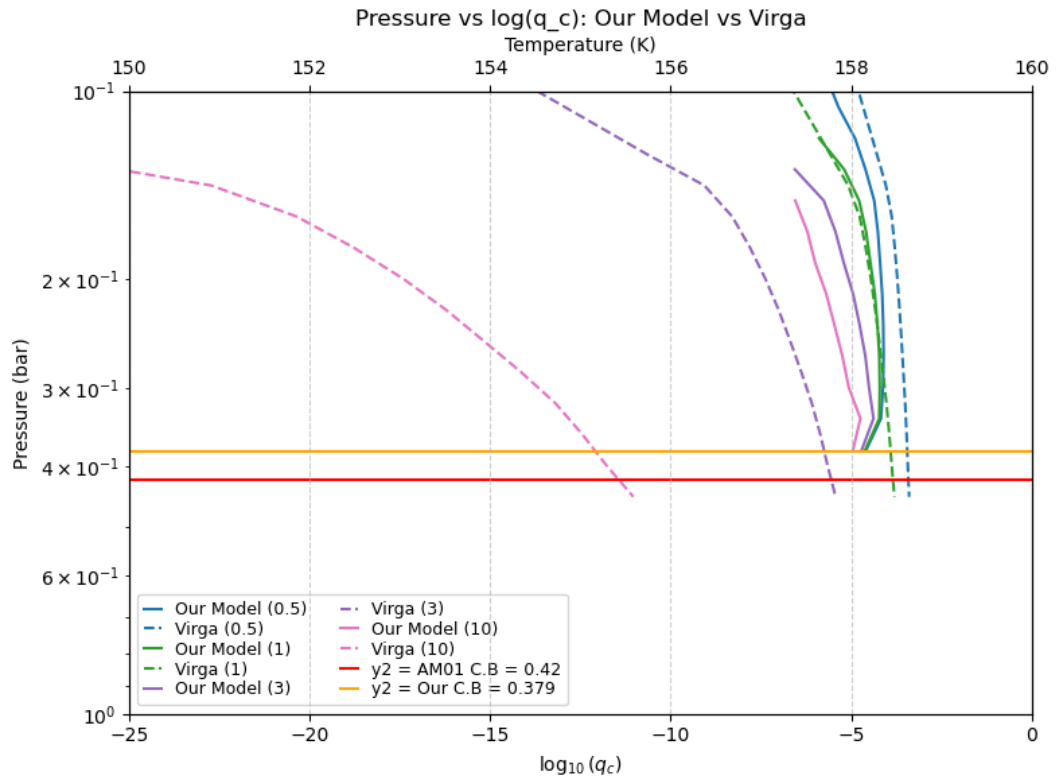


Figure B.2: Pressure vs $\log_{10}q_c$ for various f_{rain} values and compared with 1.1 of my model and [5]

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